Unit: 5 Unit 9: Data Statistical Analysis **Name:** ____________
Topic: Box Whisker (Dispersion) **Date:** ________

Foundation (Jalapeno)

Q1. What is the primary purpose of a box-and-whisker plot?
(A) To display the mean of a dataset
(B) To show the dispersion of a dataset
(C) To compare the median of two datasets
(D) To identify the mode of a dataset
(E) To calculate the range of a dataset

Q2. Which part of a box-and-whisker plot represents the interquartile range?
(A) The whiskers
(B) The box
(C) The median line
(D) The outliers
(E) The axes

Q3. What is the term for the difference between the third quartile (Q_3) and the first quartile (Q_1)?
(A) Range
(B) Interquartile range
(C) Median
(D) Mode
(E) Standard deviation

Q4. Which of the following is a characteristic of an outlier in a box-and-whisker plot?
(A) It is located within the box
(B) It is located on the median line
(C) It is more than $1.5 \cdot IQR$ away from Q_1 or Q_3
(D) It is between Q_1 and Q_3
(E) It is the median of the dataset

Q5. What does the first quartile (Q_1) represent in a dataset?
(A) The median of the dataset
(B) The lower 25
(B) The upper 25
(B) The range of the dataset
(C) The mode of the dataset

Q6. Which of the following statements about the third quartile (Q_3) is true?
(A) It is the median of the lower half of the data
(B) It is the median of the upper half of the data
(C) It is the maximum value in the dataset
(D) It is the minimum value in the dataset
(E) It is always equal to the median

Q7. What can be inferred about a dataset if its box-and-whisker plot shows a box that is skewed to the right?
(A) The dataset is symmetric
(B) The dataset has a large number of outliers
(C) The dataset has a higher median than mean
(D) The dataset has a lower median than mean
(E) The dataset is normally distributed

Q8. What does a box-and-whisker plot use to represent the median of a dataset?
(A) A whisker
(B) The box
(C) A line inside the box
(D) A point outside the box
(E) An axis label

Open Ended Questions (FRQ)

Q9. Construct a box-and-whisker plot for the following dataset: $2, 4, 6, 8, 10, 12, 14, 16, 100$. Identify the quartiles, interquartile range, and any outliers.

Q10. Compare the box-and-whisker plots of two datasets. Dataset A: $10, 12, 14, 16, 18, 20$ and Dataset B: $5, 10, 15, 20, 25, 30$. Discuss the similarities and differences in terms of dispersion, quartiles, and outliers.

Chef's Tips

- Tip 1: Ensure students understand the construction of box-and-whisker plots before moving on to analysis.
- Tip 2: Highlight the importance of identifying quartiles and interquartile range in understanding dataset dispersion.
- Tip 3: Emphasize the role of box-and-whisker plots in detecting outliers and understanding their impact on dataset analysis.